BOUNDARY CONDITIONS FOR CONVECTIVE HEAT TRANSFER IN OpenFOAM®

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with the knowledge aquired during the development of the Final Thesis entitled **"Thermal analysis with OpenFOAM® of a neutron flux furnace in order to study the thermal gradients in the sample"**.

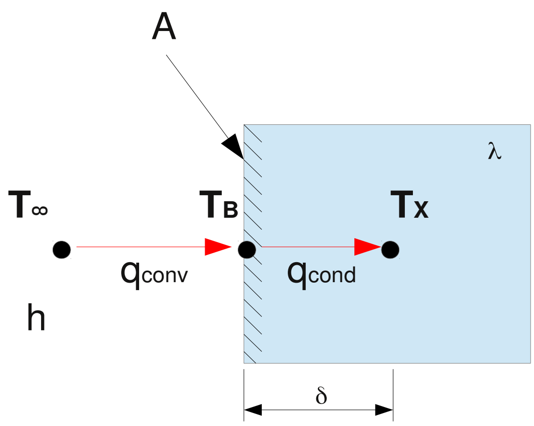
A summarized version of this thesis can be seen at **http://foamingtime.wordpress.com/**

BASIC CONVECTION

Knowing that $q_{conv}=q_{cond}$ (1)

$$q_{conv}=h\cdot A\cdot(T_\infty-T_B) \quad (2)$$

$$q_{cond}=\frac{\lambda}{\delta}\cdot A\cdot(T_B-T_X) \quad (3)$$



We get

$$\frac{q}{A}= h\cdot(T_\infty-T_B)= \frac{\lambda}{\delta}\cdot(T_B-T_X) \quad (4)$$

In OpenFOAM values at boundaries are calculated as follows:

$$X_B= f\cdot X_{ref} + (1-f)\cdot\left[X_X+\frac{Grad_{ref}(X)}{deltaCoeffs}\right] \quad (5)$$

Being X_{ref} a reference value for the variable at boundary, X_x the value of the variable in the cell center, $Grad_{ref}(X)$ the reference gradient of the variable, *deltaCoeffs* the inverse of the face-center to cell-center distance $\left(deltaCoeffs=\frac{1}{\delta}\right)$ and *f* a weighted factor that defines the BC type, beeing a Dirichlet BC when *f*=1, a Neumann BC when *f*=0 and a Robin BC when 1>*f*>0.

Eq. (4) has to be converted into something similar to Eq. (5), so let's do it! First we have to isolate T_B from Eq. (4)

$$\left(\frac{\lambda}{\delta}+h\right)\cdot T_B= h\cdot T_\infty + \frac{\lambda}{\delta}\cdot T_X \quad \rightarrow \quad T_B=\frac{h}{\left(\frac{\lambda}{\delta}+h\right)}\cdot T_\infty + \frac{\frac{\lambda}{\delta}}{\left(\frac{\lambda}{\delta}+h\right)}\cdot T_X$$

If we rename some terms of the equation

$$f=\frac{h}{\left(\frac{\lambda}{\delta}+h\right)} \quad ; \quad 1-f=\frac{\frac{\lambda}{\delta}}{\left(\frac{\lambda}{\delta}+h\right)}$$

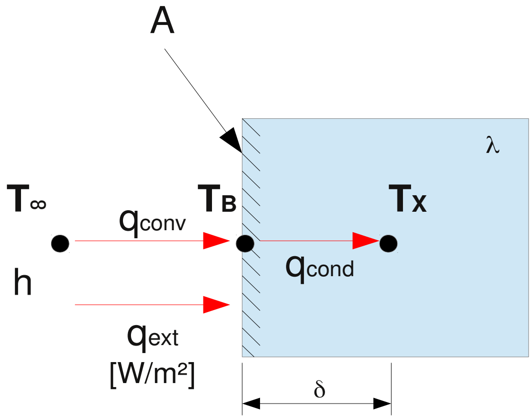
Thus, we obtain

$T_B= f\cdot T_\infty + (1-f)\cdot T_X$ being T_∞ the **reference value**.

We can even rewrite *f* in a better looking entity $f=\frac{1}{1+\frac{\lambda}{h\cdot\delta}}$

CONVECTION WITH ADDITION OF A CONSTANT HEAT FLUX

Now the balance satisfies that



$$q_{conv}+q_{ext}=q_{cond}$$

Thus,

$$\frac{q}{A}= h\cdot(T_{\infty}-T_B) + q_{ext}= \frac{\lambda}{\delta}\cdot(T_B-T_X)$$

As it was said before, T_B must be isolated,

$$\left(\frac{\lambda}{\delta}+h\right)\cdot T_B= h\cdot T_{\infty} + \frac{\lambda}{\delta}\cdot T_X + q_{ext}$$

$$\downarrow$$

$$T_B= \frac{h}{\left(\frac{\lambda}{\delta}+h\right)}\cdot T_{\infty} + \frac{\frac{\lambda}{\delta}}{\left(\frac{\lambda}{\delta}+h\right)}\cdot T_X + \frac{q_{ext}}{\left(\frac{\lambda}{\delta}+h\right)}$$

$$\downarrow$$

$$T_B= \frac{h}{\left(\frac{\lambda}{\delta}+h\right)}\cdot T_{\infty} + \frac{\frac{\lambda}{\delta}}{\left(\frac{\lambda}{\delta}+h\right)}\cdot T_X + \frac{h}{\left(\frac{\lambda}{\delta}+h\right)}\cdot\frac{q_{ext}}{h} \rightarrow \boxed{T_B= \frac{h}{\left(\frac{\lambda}{\delta}+h\right)}\cdot\left(T_{\infty}+\frac{q_{ext}}{h}\right) + \frac{\frac{\lambda}{\delta}}{\left(\frac{\lambda}{\delta}+h\right)}\cdot T_X}$$

Again $\boxed{f=\frac{1}{1+\frac{\lambda}{h\cdot\delta}}}$ but this time the **reference value** is $\boldsymbol{T_{\infty}+\frac{q_{ext}}{h}}$.

Note: By doing another trick it could have been obtained a **reference value** of $\boldsymbol{T_{\infty}}$ and a **reference gradient** of $\boldsymbol{\frac{q_{ext}}{\lambda}}$ instead.

CONVECTION WITH ADDITION OF A CONSTANT HEAT FLUX AND AN EXTRA EXTERNAL SOLID LAYER

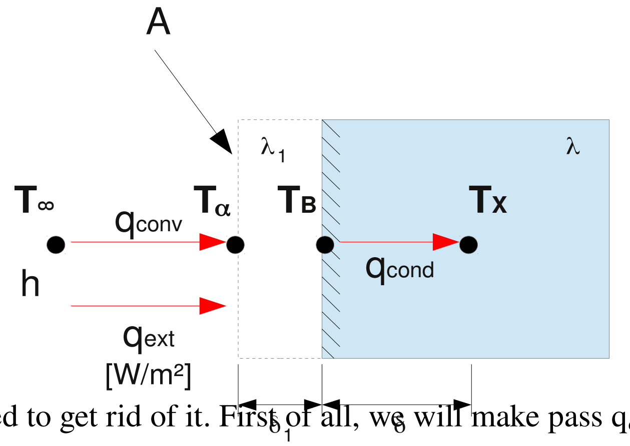
Remember that the external solid layer <u>does not belong</u> to the calculation domain!

In this case

$$q_{conv}+q_{ext}=q_{cond1}=q_{cond2}$$

Therefore,

$$\frac{q}{A}=h\cdot(T_{\infty}-T_{\alpha})+q_{ext}=\frac{\lambda_1}{\delta_1}\cdot(T_{\alpha}-T_B)=$$
$$=\frac{\lambda}{\delta}\cdot(T_B-T_X)$$



T_{α} is a variable that we cannot manage, so we need to get rid of it. First of all, we will make pass q_{ext} within the brackets.

$$\frac{q}{A}=h\cdot\left(T_{\infty}-T_{\alpha}+\frac{q_{ext}}{h}\right)=h\cdot\left(\left(T_{\infty}+\frac{q_{ext}}{h}\right)-T_{\alpha}\right)=\frac{\lambda_1}{\delta_1}\cdot(T_{\alpha}-T_B)=\frac{\lambda}{\delta}\cdot(T_B-T_X)$$

As it is known,

$$q=\frac{T_A-T_B}{R_{AB}}=\frac{T_B-T_C}{R_{BC}}\rightarrow q=\frac{T_A-T_C}{R_{AC}}\quad with\quad R_{AC}=R_{AB}+R_{BC}$$

Therefore,

$$\frac{q}{A}=\frac{\left(T_{\infty}+\frac{q_{ext}}{h}\right)-T_B}{\frac{1}{h}+\frac{\delta_1}{\lambda_1}}=\frac{\lambda}{\delta}\cdot(T_B-T_X)$$

Again, T_B has to be isolated, let's do it!

$$T_{\infty}+\frac{q_{ext}}{h}-T_B=\frac{\lambda}{\delta}\cdot\frac{\lambda_1+h\cdot\delta_1}{\lambda_1\cdot h}\cdot(T_B-T_X)$$

To make it cleaner,

$$\beta=\frac{\lambda}{\delta}\cdot\frac{\lambda_1+h\cdot\delta_1}{\lambda_1\cdot h}$$

Thus,

$$T_\infty + \frac{q_{ext}}{h} - T_B = \beta \cdot (T_B - T_X) \quad \rightarrow \quad T_B \cdot (1+\beta) = T_\infty + \frac{q_{ext}}{h} + \beta \cdot T_X \quad \rightarrow$$

$$\rightarrow \quad \boxed{T_B = \frac{1}{1+\beta} \cdot \left(T_\infty + \frac{q_{ext}}{h} \right) + \frac{\beta}{1+\beta} \cdot T_X}$$

Hence it is deduced that this is the same case than the previous one, except for the value of *f*, which now it has a value of $f = \frac{1}{1+\beta}$ being $\beta = \frac{\lambda}{\delta} \cdot \frac{\lambda_1 + h \cdot \delta_1}{\lambda_1 \cdot h}$.

Summary

This is the general expression for a boundary condition:

$$\boxed{X_B = f \cdot refValue + (1-f) \cdot \left[X_X + \frac{refGrad}{deltaCoeffs} \right]}$$

The following table shows the values that each term has to take for every one of the 3 modes of convection studied.

Convection type	**f**	**refValue**		**refGrad**		**β**
		OPTION 1	OPT. 2	OPT. 1	OPTION 2	
Basic	$\frac{1}{1+\beta}$	T_∞	-	0	-	$\frac{\lambda}{h \cdot \delta}$
Constant Heat Flux	$\frac{1}{1+\beta}$	$T_\infty + \frac{q_{ext}}{h}$	T_∞	0	$\frac{q_{ext}}{\lambda}$	$\frac{\lambda}{h \cdot \delta}$
Heat Flux + Extra Layer	$\frac{1}{1+\beta}$	$T_\infty + \frac{q_{ext}}{h}$	T_∞	0	$\frac{q_{ext}}{\lambda} \cdot \frac{\lambda_1}{\lambda_1 + h \cdot \delta_1}$	$\frac{\lambda}{h \cdot \delta} \cdot \frac{\lambda_1 + h \cdot \delta_1}{\lambda_1}$